A Lending Fairness Audit: What We Did, What We Found

Siddhant Jain
Simon Fraser University
sja178@sfu.ca

Ayaan Lalani
Simon Fraser University

1 WHAT WE DID

A fairness audit over three lending decisions — consumer credit (German Credit), mortgage origination (HMDA Georgia), peer-to-peer default risk (Lending Club). A deterministic AIF360 layer computes every metric and assigns every severity label by fixed rule; an LLM supplies only interpretation and is scored against that ground truth. Audits run on the **full cleaned population** (1,000 and 10,978 rows), not a test split: every row is scored out-of-fold by a model that never saw it. Frozen prompts, a no-key replay path, two anti-fabrication detectors, a $15 cumulative cap. **Spend: $0.8933 over 98 calls.** 60 tests, CI green.

2 FIVE FINDINGS

1. The reporting floor erases the largest disparities. An $n \geq 15$ floor suppressed **5 of 11** HMDA `race`×`sex` cells on a test split. Three were worse than anything reported: the audit returned MODERATE at 0.8169 while the worst real disparity was **0.1917** — wrong group, wrong by two severity bands. At full population 1 of 12 is suppressed and the worst cell (0.6734) becomes visible. Every suppressed cell belonged to a racial minority.

2. The age audit used the wrong statute. 40+ is the ADEA, an employment threshold; ECOA protects 62+. Pooling young against senior applicants cancels them, so banded `age_group` reads 0.9791 (LOW) while the statutory cut reads **0.8897** — the largest age disparity in the data. The `applicant_age_above_62` flag was in the raw file, unused.

3. Metric orientation inverted a headline number. UC3 `gender` read DI = 0.0 (CRITICAL) because selection rates were computed on the unfavorable label of a target named `loan_default`. Correct value: 0.9931. Correct toolkit, clean data, sound metric, maximally wrong conclusion.

4. Our own detectors penalised correct answers. A refusal detector flagged 6 responses, *none* a refusal (any field under 40 characters counts as empty, under a prompt demanding terseness). On a second provider it mechanically zeroed 14 cycles, moving the mean from **70.0** to **19.79** — and a *permanent block* was written into our config on that evidence. Separately, a fabricated-metric detector flagged `gpt-5-mini` 5× more often than `gpt-4o`, but **67%** of those flags are values derivable from the context (88.88 is 0.8888 as a percentage). The better-reasoning model is penalised for showing its work.

5. Three of our own published findings do not survive. A CRITICAL intersectional ratio of 0.6087 becomes **0.8201** and clears four-fifths once its *15-person, 15/15* comparator grows to 69. An inference about label bias came from a model with target leakage (`interest_rate`: 0.0% missing on approvals, 99.3% on denials; AUC 0.9942 → **0.7915** once removed). And "0.8956 fails the 0.80 screen" is arithmetically false. UC3's `gender` is also synthetic (`np.random.random` > 0.5), so UC3 contains no protected class.

3 CROSS-MODEL BENCHMARK

Same corrected packs, both models. **Prompt sensitivity is model-generation-specific**: cycle-mean spread across four prompt strategies is 14.7 / 9.0 points for `gpt-4o` (german_credit / hmda) and **0.0** / **1.9** for `gpt-5-mini`. The project's earlier "prompt phrasing moves the verdict" result reproduces on the older model and nearly vanishes on the newer one. `gpt-5-mini` rejects `temperature` outright, so a temperature sweep is impossible on it.

4 ARTIFACTS

Results `artifacts/consolidated/RESULTS.md`; full draft `report/report_facct.tex`; benchmark-framed variant `report/report.tex`; population comparison `scripts/compare_populations.py`.

Table 1: Marginal disparate impact, full population. †The ECOA cut (Reg B §1002.2(o) defines elderly as 62+); `age_group` is the banded view. ‡Synthetic attribute — see Finding 5.

Case	Attribute	DI
UC1	`Sex`	0.8888
UC1	`AgeGroup`	0.9258
UC1	`foreign_worker`	0.8387
UC2	`race`	0.9371
UC2	`sex`	0.9572
UC2	`age_group` (banded)	0.9791
UC2	`age_62_plus`†	**0.8897**
UC3	`gender`‡	0.9931